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Inventor(s)	Yu; Chia-Ta et al.

Epitaxial layer under a gate structure of a transistor

Abstract

The present disclosure is directed to methods for the fabrication of buried layers in gate-all-around (GAA) transistor structures to suppress junction leakage. In some embodiments, the method includes forming a doped epitaxial layer on a substrate, forming a stack of alternating first and second nano-sheet layers on the epitaxial layer, and patterning the stack and the epitaxial layer to form a fin structure. The method includes forming a sacrificial gate structure on the fin structure, removing portions of the fin structure not covered by the sacrificial gate structure, and etching portions of the first nano-sheet layers. Additionally, the method includes forming spacer structures on the etched portions of the first nano-sheet layers and forming source/drain (S/D) epitaxial structures on the epitaxial layer abutting the second nano-sheet layers. The method further includes removing the sacrificial gate structure, removing the first nano-sheet layers, and forming a gate structure around the second nano-sheet layers.

Inventors: Yu; Chia-Ta (New Taipei, TW), Huang; Yen-Chieh (Tainan, TW), Tu; Yi-Hsien (Hsinchu, TW), Wong; I-Hsieh (Hsinchu, TW)

Applicant: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

Family ID: 1000008779800

Assignee: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9093530	12/2014	Huang et al.	N/A	N/A
9171929	12/2014	Lee et al.	N/A	N/A
9214555	12/2014	Oxland et al.	N/A	N/A
9236267	12/2015	De et al.	N/A	N/A
9520482	12/2015	Chang et al.	N/A	N/A
9548303	12/2016	Lee et al.	N/A	N/A
9564489	12/2016	Yeo et al.	N/A	N/A
9576814	12/2016	Wu et al.	N/A	N/A
9601342	12/2016	Lee et al.	N/A	N/A
9608116	12/2016	Ching et al.	N/A	N/A
10453824	12/2018	Mochizuki et al.	N/A	N/A
11901415	12/2023	Yu et al.	N/A	N/A
2015/0295085	12/2014	Yu	438/197	H01L 29/1083
2019/0103317	12/2018	Yu	N/A	H01L 29/0673
2020/0144367	12/2019	Tsou	N/A	H01L 29/785
2020/0273979	12/2019	Xie	N/A	B82Y 10/00

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-provisional patent application Ser. No. 17/333,276, titled “Transistor Isolation Structures,” filed on May 28, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND

(1) Source/drain regions in fin-based field effect transistors (finFETs) are grown from side surfaces of the fin structures and a top surface of the semiconductor substrate on which the fin structures are formed. During operation, a leakage current path can be formed between the source/drain regions and the semiconductor substrate. The leakage current can degrade the finFET's performance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures.

(2) FIG. 1 is a cross-sectional view of gate-all-around nano-sheet FETs formed on a buried layer disposed on a substrate, in accordance with some embodiments.

(3) FIG. 2 is a cross-sectional view of gate-all-around nano-sheet FETs formed on a substrate, in accordance with some embodiments.

(4) FIGS. 3A and 3B are flow charts of a method for the fabrication of gate-all-around nano-sheet FETs on a buried layer disposed on a substrate, in accordance with some embodiments.

(5) FIGS. 4 and 5 are cross-sectional views of intermediate structures during the fabrication of gate-all-around nano-sheet FETs on a buried layer disposed on a substrate, in accordance with some embodiments.

(6) FIG. 6 is an isometric view of an intermediate structure during the fabrication of gate-all-around nano-sheet FETs on a buried layer disposed on a substrate, in accordance with some embodiments.

(7) FIGS. 7-12 are cross-sectional views of intermediate structures during the fabrication of gate-all-around nano-sheet FETs on a buried layer disposed on a substrate, in accordance with some embodiments.

(8) FIGS. 13A-13D are cross-sectional views of n-type and p-type gate-all-around nano-sheet FETs with different types of buried layers on a substrate, in accordance with some embodiments.

DETAILED DESCRIPTION

(9) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed that are between the first and second features, such that the first and second features are not in direct contact.

(10) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship

to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(11) In some embodiments, the terms “about” and “substantially” can indicate a value of a given quantity that varies within 5% of the value (e.g., $\pm 1\%$, 2% , 3% , $\pm 4\%$, $\pm 5\%$ of the value). These values are merely examples and are not intended to be limiting. It is to be understood that the terms “about” and “substantially” can refer to a percentage of the values as interpreted by those skilled in relevant art(s) in light of the teachings herein.

(12) Gate-all-around (GAA) field effect transistors (GAA-FETs), such as nano-sheet or nano-wire GAA-FETs, have improved gate control over their channel regions compared to other types of FETs whose gate structure covers sidewall portions and top surfaces of semiconductor fin structures. Due to their gate-all-around geometry, GAA nano-sheet or nano-wire FETs achieve larger effective channel widths and higher drive currents. At the same time, their distinct geometry makes GAA nano-sheet or nano-wire FETs susceptible to leakage current. For example, the gate electrode, which surrounds the nano-sheets of the FET, is formed in close proximity to the semiconductor substrate. As a result, and during operation, a parasitic channel can be formed within the semiconductor substrate between the source/drain terminals. This parasitic channel can degrade the performance of the GAA FET and increase the transistor's power consumption. To suppress the parasitic channel formation, the semiconductor substrate can be “counter-doped” with implants—e.g., the semiconductor substrate is doped with a dopant type opposite to the dopant type in the channel region of the nano-sheet or nano-wires. However, doping the semiconductor substrate with an implant process adds cost to the manufacturing process and/or may not effectively eliminate or suppress the parasitic channel formation.

(13) In addition to the above, etching processes used during the GAA-FET fabrication process can have undesirable “side effects,” which exacerbate the appearance of leakage current. For example, etching processes used in the formation of the channel region and the spacer structures between the gate structure and the S/D terminals of a GAA-FET can result in the formation of over-etched areas within the substrate below the S/D terminals. These over-etched areas can be susceptible to junction leakage current between the S/D terminals and the substrate.

(14) The embodiments described herein are directed to methods for the fabrication of buried layers in GAA FETs, which prevent the formation of over-etched areas in the substrate by subsequent etching processes. In some embodiments, the buried layer is formed between the GAA FETs and the substrate at a thickness between about 5 nm and about 20 nm. According to some embodiments, the buried layers can be formed for p-type and n-type GAA-FETs and can include doped or undoped (e.g., intrinsic) silicon layers. In some embodiments, the buried layers include a wide bandgap material, such as carbon-doped silicon. In some embodiments, the dopant selection for the silicon layers is based on the type of GAA-FETs—e.g., n-type or p-type. In some embodiments, the dopant concentration is larger than about 1×10^{21} atoms/cm³.

(15) According to some embodiments, FIG. 1 is a cross-sectional view of two GAA-FETs, **100** and **105**, formed on a buried layer **110**, which in turn is disposed on substrate **115**. Each of GAA-FETs **100** and **105** includes a gate structure **120**, which surrounds semiconductor nano-sheet (NS) or nano-wire (NW) layers **125**. By way of example and not limitation, each gate structure **120** includes a dielectric stack formed by an interfacial dielectric layer **120a** and a high-k dielectric **120b**. Further, each gate structure **120** includes a gate electrode **120c** with capping layers, one or more work function metallic layers, and a metal fill layer not shown in FIG. 1 for simplicity.

(16) Further, GAA-FETs **100** and **105** include source/drain (S/D) epitaxial structures **130** in contact with NS or NW layers **125**. Each S/D epitaxial structure **130** includes one or more doped epitaxial layers, which are successively formed on buried layer **110**. In some embodiments, S/D epitaxial

structures **130** are electrically isolated from gate structures **120** via spacer structures **135**. Spacer structures **135** further separate NS or NW layers **125** in the vertical direction (e.g., along the z-direction) as shown in FIG. **1**.

(17) In some embodiments, sidewall surfaces of gate structures **120** in GAA-FETs **100** and **105** are covered by gate spacers **140**, which provide structural support during the formation of gate structures **120**. In addition, gate spacers **140** provide to gate structures **120** electrical isolation and protection during the formation of S/D contacts, which are not shown in FIG. **1**.

(18) In some embodiments, GAA-FETs **100** and **105** can be either n-type or p-type (e.g., pFETs or nFETs). Additional GAA-FETs, similar to the GAA-FETs shown in FIG. **1**, can be formed over substrate **115**. These additional GAA-FETs are within the spirit and the scope of this disclosure. In some embodiments, p-type GAA FETs have S/D epitaxial structures **130** with boron-doped (B-doped) silicon-germanium (SiGe) layers, B-doped germanium (Ge) layers, B-doped germanium-tin (GeSn) layers, or combinations thereof. In some embodiments, n-type GAA FETs have S/D epitaxial structures **130** with arsenic (As)-doped silicon layers (Si—As), phosphorous (P)-doped silicon layers (Si—P), carbon-containing silicon layers (Si—C), or combinations thereof. In some embodiments, S/D epitaxial structures **130** are grown directly on top surfaces of buried layer **110**.

(19) In some embodiments, buried layer **110** has a thickness between about 5 nm and about 20 nm, and includes a doped silicon layer that forms a p-n junction depletion region between GAA-FETs **100/105** and substrate **115** to suppress any junction leakage current between GAA-FETs **100/105** to substrate **115**. In some embodiments, dopants in buried layer **110** are introduced during the growth process as opposed to being introduced with an ion implantation process after growth. Introduction of dopants during growth offers benefits over ion implantation. For example, the dopant concentration achieved during growth can be higher than that achieved by ion implantation. In some embodiments, the dopant concentration achieved during the growth can be greater than about 1×10^{21} atoms/cm³. Further, the dopants can be uniformly distributed across the thickness of buried layer **110** as opposed to being normally distributed within buried layer **110**. This means that dopants introduced during growth can have a uniform dopant profile across the thickness of buried layer **110** compared to implanted dopants. In some embodiments, an activation anneal is performed after growth.

(20) In some embodiments, buried layer **110** can be doped with As, P, or combinations thereof when GAA FETs **100** and **105** are p-type transistors and with B when GAA FETs **100** and **105** are n-type transistors. This combination of dopants ensures that a p-n junction depletion region is formed between GAA-FETs **100/105** and substrate **115** to suppress the flow of leakage current from GAA-FETs **100/105** to substrate **115**.

(21) In some embodiments, buried layer **110** can be a silicon layer containing C at a concentration of less than about 5 atomic percent (at. %) to form a Si—C wide bandgap barrier between GAA-FETs **100/105** and substrate **115**. In some embodiments, the Si—C bandgap is between about 3 eV and 3.4 eV, which is wider than the intrinsic silicon bandgap of about 1.12 eV. In some embodiments, Si—C buried layers **110** are applicable to both n-type and p-type GAA-FETs. This is because Si—C forms an insulating barrier between GAA-FETs **100/105** and substrate **115** as opposed to a p-n junction depletion region, whose dopants selection is based on the type of the transistors (n-type or p-type) formed thereon.

(22) In some embodiments, buried layer **110** acts as an etch mask that protects substrate **115** from being etched by an etching process used in the formation of the channel region and spacer structures **135**. If buried layer **110** is not present and the aforementioned etching process etches substrate **115**, S/D epitaxial structures **130** will be formed within the etched areas of substrate **115**. Therefore, during operation of GAA FETs **100** and **105**, a leakage current path can be formed between GAA-FETs **100/105** and substrate **105**. By way of example and not limitation, and in referring to FIG. **2**, arrows **200** represent the leakage current path between GAA-FETs **100/105** and substrate **105** when buried layer **110** is not present and an etching process etches exposed portions

of substrate **115**. In this case, substrate **115** can be counter-doped with an ion implant process to limit the amount of leakage current. However, ion implant processes add complexity to the fabrication process and increase the fabrication cost.

(23) According to some embodiments, FIGS. **3A** and **3B** are flow charts of a fabrication method **300** for the formation of buried layer **110** below GAA-FETs **100** and **105** shown in FIG. **1**. Other fabrication operations can be performed between the various operations of method **300** and are omitted merely for clarity. This disclosure is not limited to this operational description. Rather, other operations are within the spirit and scope of the present disclosure. It is to be appreciated that additional operations may be performed. Moreover, not all operations may be needed to perform the disclosure provided herein. Additionally, some of the operations may be performed simultaneously, or in a different order than the ones shown in FIGS. **3A** and **3B**. In some embodiments, one or more other operations may be performed in addition to or in place of the presently described operations. For illustrative purposes, method **300** is described with reference to the embodiments shown in FIGS. **4** through **12**.

(24) In referring to FIG. **3A**, method **300** begins with operation **305** and the process of forming a buried layer (e.g., buried layer **110**) on a substrate (e.g., substrate **115**). In some embodiments, the substrate (e.g., substrate **115**) includes silicon or another elementary semiconductor, such as germanium (Ge). In some embodiments, the substrate includes a compound semiconductor like silicon carbide (SiC), gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), or indium antimonide (InSb). In some embodiments, the substrate includes an alloy semiconductor like silicon germanium (SiGe), gallium arsenide phosphide (GaAsP), aluminum indium arsenide (AlInAs), aluminum gallium arsenide (AlGaAs), gallium indium arsenide (GaInAs), gallium indium phosphide (GaInP), or gallium indium arsenide phosphide (GaInAsP). In some embodiments, the substrate includes any combination of the aforementioned materials. In some embodiments, substrate **115** is lightly doped (e.g., P-doped) with a dopant concentration between about 1×10^{14} atoms/cm³ and about 1×10^{15} atoms/cm³.

(25) In some embodiments, the buried layer is grown on top surfaces of the substrate. The buried layer can be epitaxially grown directly on the substrate with a chemical vapor deposition (CVD) process with silicon precursors, such as silane (SiH₄), disilane (Si₂H₆), dichlorosilane (SiH₂Cl₂), other suitable gases, or combinations thereof. During growth, dopants (e.g., B, P, As) are introduced in the buried layer with appropriate precursor gases, such as diborane (B₂H₆), phosphine (PH₃), arsine (AsH₃), other suitable gases, or combinations thereof. In some embodiments, the dopant concentration in the buried layer is set to be higher than about 1×10^{21} atoms/cm³ to ensure that a p-n junction with a sufficient depletion width is formed between the GAA FET and the buried layer to prevent junction leakage current. As discussed above, incorporation of dopants during the epitaxial growth ensures a uniform distribution of dopants across the thickness of the buried layer and a high dopant concentration (e.g., compared to ion implantation). In some embodiments, carbon is incorporated to buried layer **110** with the addition of organic gases, such as monomethylsilane (MMS). The incorporated amount of carbon in buried layer **110** is less than about 5 at. % according to some embodiments. Carbon increases the bandgap of buried layer and reduces its current conduction ability. In some embodiments, carbon concentrations higher than about 5 at. % introduce mechanical stress to the transistors, which is undesirable because it impacts the transistor's electrical characteristics and induces unwanted variability between transistors. Additionally, carbon concentrations higher than about 5 at. % require higher thermal budgets, which increase the manufacturing cost and can be challenging to integrate into the manufacturing flow.

(26) In some embodiments, the buried layer can be deposited at a temperature between about 550° C. and 800° C. and at a process pressure between about 1 Torr and about 600 Torr.

(27) In some embodiments, the buried layer is deposited at a thickness between about 5 nm and about 20 nm. In some embodiments, thicknesses below about 5 nm do not provide sufficient

protection against subsequent etching operations. On the other hand, thicknesses greater than 20 nm can induce mechanical stress to the GAA FETs formed thereon, which can alter the electrical characteristics of the GAA-FET. In some embodiments, the as-deposited thickness of buried layer can be adjusted based on the etching rate of buried layer by the etching process used in subsequent operations. For example, an etching process with a high etching rate requires a thicker as-deposited buried layer compared to an etching process with a lower etching rate. The additional thickness compensates for any material loss experienced by the buried layer during the etching process so that the thickness of the barrier layer after the etching is no less than about 5 nm.

(28) According to some embodiments, FIG. 4 is a cross-sectional view of substrate 115 after the deposition of buried layer 110 at a thickness $110t$ according to operation 305 of method 300. By way of example and not limitation, buried layer 110 can be deposited to cover the entire top surface of substrate 115. In some embodiments, barrier layer 110 can be patterned to cover selective areas of substrate 115 where GAA-FETs are formed—for example, buried layer 110 is removed from areas of substrate 115 where GAA-FETs are not formed.

(29) In referring to FIG. 3A, method 300 continues with operation 310 and the process of forming a stack of alternating first and second NS layers on buried layer 110. In some embodiments, the formation of the stack according to operation 310 includes a sequential deposition of NS layer 125, shown in FIG. 1, and another type of NS layer different from NS layer 125. By way of example and not limitation, FIG. 5 shows a stack 500 of alternating NS layer 505 (first NS layer) and NS layer 125 (second NS layer) formed on buried layer 110 according to operation 310. In some embodiments, the material of NS layers 505 (first NS layers) in stack 500 is selected so that NS layers 505 can be selectively removed via etching from stack 500 without removing NS layers 125 (second NS layers). For example, if NS layers 125 are silicon (Si) NS layers, NS layers 505 can be silicon germanium (SiGe) NS layers. In some embodiments, the deposition sequence starts with the type of NS layers to be removed; for example, in this case with NS layer 505 (first NS layer).

(30) In some embodiments, layers 125 and 505 are referred to as “nano-sheets” when their width along the y-direction is substantially different from their height along z-direction (e.g., when the width is larger/narrower than their height). In some embodiments, layers 125 and 505 are referred to as “nano-wires” when their width along the y-direction is equal to their height along z-direction. In some embodiments, layers 125 and 505 are deposited as nano-sheets and subsequently patterned to form nano-wires with equal height and width. By way of example and not limitation, layers 505 and 125 will be described in the context of nano-sheets (NS) layers (e.g., first and second NS layers, respectively). Based on the disclosure herein, nano-wires, as discussed above, are within the spirit and the scope of this disclosure. Further, for example purposes and without limiting the scope of this disclosure, NS layers 125 and 505 in method 300 will be described in the context of Si and SiGe NS layers.

(31) NS layers 125 and 505 can be grown using any suitable method. For example, NS layers 125 and 505 can be grown using a CVD process with precursor gases like $\text{SiH}_{3.4}$, $\text{Si}_{2.6}\text{H}_2$, $\text{Si}_{2.2}\text{Cl}_{0.2}$, germane (GeH_4), digermane ($\text{Ge}_{2.6}\text{H}_2$), other suitable gases, or combinations thereof. In some embodiments, NS layers 505 include between about 20 at. % and about 30 at. % Ge while NS layers 125 are substantially germanium-free—e.g., have a Ge concentration between about 0.1% and about 1%. In some embodiments, NS layers 125, which form the channel region of GAA-FET 100, can be lightly doped or intrinsic (e.g., un-doped). If lightly doped, the doping level of NS layers 125 is less than about 10^{13} atoms/cm³. NS layers 505 and 125 can be deposited sequential without a vacuum break (e.g., in-situ) to avoid the formation of any intervening layers. In some embodiments, NS layers 505 can be doped to increase their etching selectivity compared to NS layers 125 in a subsequent etching operation.

(32) In some embodiments, the thickness of NS layers 505 controls the spacing between every other NS layer 125 in stack 500. The thickness of each NS layer 125 and 505 can range, for example, from about 5 nm to about 8 nm. Since the NS layers 125 and 505 are grown individually,

the thickness of each NS layer **505** and **125** can be adjusted independently based, for example, on the deposition time.

(33) In referring to FIG. **3A**, method **300** continues with operation **315** and the process of patterning stack **500** to form fin structures. In some embodiments, stack **500** is patterned to form fin structures with a width along the y-direction and a length along the x-direction. The fin structures can be formed by patterning with any suitable method. For example, the fin structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Double-patterning or multi-patterning processes can combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in some embodiments, a sacrificial layer is formed over stack **500** and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used as masking structures to pattern the fin structures.

(34) By way of example and not limitation, FIG. **6** is an isometric view of fin structures **510** formed from stack **500** with the aforementioned patterning process. Additional fin structures, like fin structures **510**, can be formed on substrate **115** in the same or different area of substrate **115**. These additional fin structures are not shown in FIG. **6** for simplicity. By way of example and not limitation, each fin structure **510** has a width along the y-direction between about 15 nm and about 150 nm.

(35) In some embodiments, after the formation of fin structures **510**, an isolation structure **600** is formed on substrate **115** to cover sidewall surfaces of buried layer **110**. In some embodiments, isolation structure **600** is a shallow trench isolation (STI) structure that electrically isolates fin structures **510** and includes a silicon oxide based dielectric. By way of example and not limitation, isolation structure **600** can be formed as follows. Isolation structure material (e.g., a silicon oxide based dielectric) is blanket deposited over fin structures **510** and substrate **115**. The as-deposited isolation structure material is planarized (e.g., with a chemical mechanical polishing (CMP) process) so that the top surface of the isolation structure material is coplanar with the top surface of fin structures **510**. The planarized isolation structure material is etched-back so that the resulting isolation structure **600** has a height substantially similar to buried layer **110**, as shown in FIG. **6**. In some embodiments, stack **500** protrudes from isolation structure **600** so that isolation structure **600** does not cover any sidewall portion of stack **500** as shown in FIG. **6**. This is intentional and facilitates the formation of GAA-FETs **100** and **105** shown in FIG. **1**.

(36) In referring to FIG. **3A**, method **300** continues with operation **320** and the process of forming sacrificial gate structures on fin structures **510**. In some embodiments, the sacrificial gate structures are formed with their length along the y-direction—e.g., perpendicular to fin structures **510** shown in the isometric view of FIG. **6**—and their width along the x-direction. By way of example and not limitation, FIG. **7** is a cross-sectional view of FIG. **6** along cut-line AB. FIG. **7** shows sacrificial gate structures **700** formed on portions of fin structures **510**. Because FIG. **7** is a cross-sectional view, portions of gate structures **700** covering sidewall portions of fin structures **510** are not shown. Further, in the cross-sectional view of FIG. **7**, only one of fin structures **510** from FIG. **6** is shown. In some embodiments, portions of gate structures **700** are formed between fin structures **510** and on isolation structure **600** shown in FIG. **6**.

(37) In some embodiments, gate structures **700** cover top and sidewall portions of fin structures **510**. In some embodiments, sacrificial gate structures **700** are replaced by gate structures **120** shown in FIG. **1** during a gate replacement process. Gate structures **700** include a sacrificial gate electrode **700a** and a sacrificial gate dielectric not shown in FIG. **7** for simplicity. At this fabrication stage, gate spacers **140** and capping layers **705** are formed on sidewall and top surfaces of sacrificial gate structures **700**. As discussed above, gate spacers **140** are not removed during the gate replacement process; instead, gate spacers **140** facilitate the formation of gate structures **120**.

Capping layers **705** protect sacrificial gate electrode **700a** from subsequent etching operations discussed below.

(38) By way of example and not limitation, sacrificial gate structures **700** are formed by depositing and patterning the sacrificial gate dielectric and gate electrode layers over fin structures **510**. Gate spacers **140** and capping layer **705** are formed once sacrificial gate structures **700** are formed. In some embodiments, sacrificial gate structures **700** are formed over multiple fin structures **510**. As shown in FIG. 7, edge portions of fin structures **510** are not covered by sacrificial gate structures **700**. This is because, the width of sacrificial gate structures **700** is narrower than the length of fin structures **510** along the x-direction. In some embodiments, sacrificial gate structures **700** are used as masking structures in subsequent etching operations to define the channel region of GAA-FETs **100** and **105** shown in FIG. 1. For this reason, the lateral dimensions (e.g., the width and length) of sacrificial gate structures **700** and gate structures **120** are substantially similar.

(39) In referring to FIG. 3B, method **300** continues with operation **325** and the process of removing (e.g., “trimming”) portions of fin structures **510** not covered by sacrificial gate structures **700**. As discussed above, the edge portions of fin structures **510** are not covered by sacrificial gate structures **700** and are therefore removed in operation **325**. In some embodiments, FIG. 8 shows the structure of FIG. 7 after operation **325**. In some embodiments, the removal process involves a dry etching process, a wet etching process, or a combination thereof. The removal process is selective towards NS layers **125** and NS layers **505** (e.g., Si and SiGe). In some embodiments, the dry etching process can include etchants having an oxygen-containing gas, a fluorine-containing gas (e.g., carbon tetrafluoride (CF₄), sulfur hexafluoride (SF₆), difluoromethane (CH₂F₂), trifluoromethane (CHF₃), and/or hexafluoroethane (C₂F₆)); a chlorine-containing gas (e.g., chlorine (Cl₂), chloroform (CHCl₃), carbon tetrachloride (CCl₄), and/or boron trichloride (BCl₃)); a bromine-containing gas (e.g., hydrogen bromide (HBr) and/or bromoform (CHBr₃)); an iodine-containing gas; other suitable etching gases and/or plasmas; or combinations thereof. The wet etching chemistry can include dilute hydrofluoric acid (DHF), potassium hydroxide (KOH) solution, ammonia; a solution containing hydrofluoric acid (HF), nitric acid (HNO₃), acetic acid (CH₃COOH); or combinations thereof.

(40) In some embodiments, the etchants of the aforementioned etching process do not substantially etch buried layer **110**. This is because the dopants in buried layer **110** reduce the etching selectivity of the etchants towards buried layer **110**. Further, the etchants do not etch sacrificial gate structures **700**—which is protected by capping layers **705** and gate spacers **140**—and isolation structure **600** shown in FIG. 6. This is because capping layers **705**, gate spacers **140**, and isolation structure **600** include materials with a low etching selectivity, such as a silicon nitride based material (e.g., silicon nitride, silicon carbon nitride, silicon carbon oxy-nitride, etc.) or silicon oxide based materials. In some embodiments, isolation structure **600** (shown in FIG. 6) and buried layer **110** are used as an etch stop layer for the etching process described above.

(41) In some embodiments, if buried layer **110** was absent from substrate **115**, the etchants used in operation **325** could recess the silicon made substrate **115** with respect to fin structures **510** to form recessed portions between the un-etched portions of each fin structure **510**. Subsequently, S/D epitaxial structures **130** formed on these recessed portions of substrate **115**, as shown in FIG. 2, would be susceptible to junction leakage current as indicated by arrows **200**. Therefore, buried layer **110** prevents the formation of recessed portions in substrate **115** and the appearance of junction leakage current between S/D epitaxial structures **130** and substrate **115**.

(42) Because of operation **325**, a “cut” is formed in each fin structure **510**. The cut divides each fin structure **510** into two separate portions; each portion covered by a sacrificial gate structure **700** as shown in FIG. 8.

(43) In some embodiments, the removal process results in a tapered profile **800** at the base of fin structure **510** as shown in FIG. 8. This can be attributed, for example, to the ability of the etchants

to reach the base of fin structures **510** in sufficient concentrations to remove portions of NS layers **505**.

(44) In referring to FIG. **3B**, method **300** continues with operation **330** and the process of selectively etching edge portions the first NS layers (NS layers **505**) from fin structures **510**. According to some embodiments, FIG. **9** shows the structure of FIG. **8** after operation **330** where exposed edges of NS layers **505** are laterally etched (e.g., recessed) along the x-direction. According to some embodiments, exposed edges of NS layers **505** are recessed (e.g., partially etched) by an amount *C* that ranges from about 2 nm to about 10 nm along the x-direction as shown in FIG. **9**. In some embodiments, spacer structures **135** shown in FIG. **1** will be formed on the recessed portions of NS layers **505**. According to some embodiments, recessed amounts less than about 2 nm result in spacer structures **135** that are too thin to provide adequate electrical isolation between gate structures **120** and S/D epitaxial structures **130** shown in FIG. **1**. On the other hand, recessed amounts greater than about 10 nm result in spacer structures **135** that reduce the gate length of the resulting transistor structures (e.g., GAA FETs **100** and **105**).

(45) In some embodiments, the recess in NS layers **505** can be achieved with a dry etching process that is selective towards SiGe. For example, halogen-based chemistries exhibit high etching selectivity towards Ge and low towards Si. Therefore, halogen gases etch Ge-containing layers (e.g., NS layers **505**) at a higher etching rate than substantially Ge-free layers (e.g., NS layers **125**). In some embodiments, the halogen-based chemistries include fluorine-based and/or chlorine-based gasses. Alternatively, a wet etching chemistry with high selectivity towards SiGe can be used. In some embodiments, a wet etching chemistry may include a mixture of sulfuric acid (H.sub.2SO.sub.4) and hydrogen peroxide (H.sub.2O.sub.2) (SPM), or a mixture of ammonia hydroxide with H.sub.2O.sub.2 and water (APM). The aforementioned etching processes are timed so that the desired amount of SiGe is removed.

(46) In some embodiments, NS layers **505** with a higher Ge atomic concentration have a higher etching rate than NS layers **505** with a lower Ge atomic concentration. Therefore, the etching rate of the aforementioned etching processes can be adjusted by modulating the Ge atomic concentration (e.g., the Ge content) in NS layers **505**. As discussed above, the Ge content in NS layers **505** can range between about 20% and about 30%. A SiGe nano-sheet layer with about 20% Ge can be etched slower than a SiGe nano-sheet layer with about 30% Ge. Consequently, the Ge concentration can be adjusted accordingly to achieve the desired etching rate and selectivity between NS layers **505** and NS layers **125**.

(47) In some embodiments, a Ge concentration below about 20% does not provide adequate selectivity between NS layers **505** and NS layers **125**. For example, the etching rate between NS layers **505** and NS layers **125** becomes substantially similar to one another and both types of nano-sheet layers can be etched during the etching process. On the other hand, for Ge concentrations higher than about 30%, Ge atoms can out-diffuse from NS layers **505** towards NS layers **125** (e.g., during growth) and change the selectivity between NS layers **505** and NS layers **125** during etching. Since Ge out-diffusion cannot be controlled, Ge concentrations higher than about 30% can result in unpredictable etching amounts.

(48) In some embodiments, the etchants used in operation **330** have a low etching selectivity towards buried layer **110**. In some embodiments, buried layer **110** is etched less than about 2 nm during operation **330**. As discussed above, the thickness of buried layer **110** can be adjusted based on the etching selectivity of the etchants used in operation **330**. For example, a thicker buried layer **110** can be used for etchants with higher etching selectivity. Similarly, a thinner buried layer **110** can be used for etchants with lower etching selectivity. In some embodiments, the low etching selectivity of buried layer **110** is achieved either because of the material included in buried layers **110** (e.g., carbon in Si:C buried layers **110**) or because of the presence of dopants (e.g., P, B, As, or combinations thereof in Si:P, Si:B, or Si:As) at high concentrations (e.g., greater than about 1×10^{21} atoms/cm³).

(49) As discussed above, if buried layer **110** was not present on substrate **115**, the etchants used in operation **325** would recess substrate **115**. These recessed portions, now exposed to the etchants of operation **330**, could be further etched and position S/D epitaxial structures **130** shown in FIG. **2** deeper into substrate **115**. The placement of S/D epitaxial structures **130** deeper into substrate **115** exacerbates junction leakage and further compromises the performance of GAA-FETs **100** and **105** shown in FIG. **2**.

(50) In referring to FIG. **3B**, method **300** continues with operation **335** and the process of forming a spacer structure (e.g., spacer structure **135**) on the etched portions of each first NS layer (e.g., each NS layer **505**). By way of example and not limitation, the formation of spacer structures **135** can be described as follows. A spacer layer **1000** is blanket deposited over the entire structure of FIG. **9** as shown in FIG. **10**. For example, spacer layer **1000** is deposited on the exposed surfaces of buried layer **110**, fin structure **510**, gate spacers **140**, and capping layer **705**. In some embodiments, spacer layer **1000** is deposited at a thickness between about 5 nm and about 10 nm to fill the recess in each NS layer **505** shown in FIG. **9**. In some embodiments, spacer layer **1000** includes a silicon-based dielectric, such as silicon nitride (SiN), silicon oxy-carbon-nitride (SiOCN), silicon carbon-nitride) SiCN, or silicon oxy-nitride (SiON). In some embodiments, spacer layer **1000** can be deposited with a plasma-enhance atomic layer deposition (PEALD) process or another suitable method capable of depositing conformal layers. As shown in FIG. **10**, spacer layer **1000** fills the space formed by the recessed edge portions of NS layers **505**. Because of the spacer layer deposition, sidewall surfaces of fin structure **510** are no longer exposed.

(51) In a subsequent operation, spacer layer **1000** is removed with an anisotropic etching process to form spacer structures **135** shown in FIGS. **1** and **11**. By way of example and not limitation, spacer layer **1000** can be removed with any dry etching chemistry capable of removing silicon nitride based materials, such as a mixture of fluorocarbon-based chemistry (e.g., fluorinated hydrocarbons) with oxygen and/or nitrogen. In some embodiments, the anisotropic etching process is terminated when spacer layer **1000** is sufficiently removed from buried layer **110** and sidewall surfaces of fin structure **510**. During the aforementioned etching process, gate spacers **140** and capping layer **705** may be partially etched.

(52) In some embodiments, the removal process results in a tapered profile **1100** at the base of fin structure **510** as shown in FIG. **11**. This can be attributed, for example, to the ability of the etchants to reach the base of fin structures **510** in sufficient concentrations to remove portions of spacer layer **1000**. For this reason, spacer structures **135** towards the base of fin structure **510** can be thicker than spacer structures **135** towards the top of fin structure **510**.

(53) In referring to FIGS. **3B** and **12**, method **300** continues with operation **340** and the process of forming S/D epitaxial structures **130** on buried layer **110**. In some embodiments, S/D epitaxial structures **130** are grown using a CVD process similar to the one used to form NS layers **505** and the NS layers **125**. For example, P-doped Si S/D epitaxial structures **130** (e.g., appropriate for n-type GAA FETs **100** and **105**) can be grown using a silane (SiH₄) precursor. The phosphorous dopant can be introduced during growth. In some embodiments, the phosphorous concentration can range from about 1×10^{21} atoms/cm³ to about 8×10^{21} atoms/cm³. The aforementioned doping concentration ranges are not limiting and other doping concentrations are within the spirit and the scope of this disclosure.

(54) Accordingly, a B-doped SiGe S/D epitaxial structures **130** (e.g., appropriate for p-type GAA FETs **100** and **105**) can include two or more epitaxial layers (not shown in FIG. **12**) grown in succession and having different Ge atomic percentages and B concentrations. For example, a first layer can have a Ge at. % that ranges from 0 to about 40%, and a B dopant concentration that ranges from about 5×10^{19} atoms/cm³ to about 1×10^{21} atoms/cm³. A second epitaxial layer can have a Ge at. % that ranges from about 20% to about 80%, and a B dopant concentration that ranges from about 3×10^{20} atoms/cm³ to about 5×10^{21} atoms/cm³. Further, a third epitaxial layer can be a capping layer that has similar Ge at. % and

B dopant concentrations with the first layer (e.g., 0 to about 40% for Ge, and about 5×10^{21} atoms/cm³ to about 1×10^{21} atoms/cm³ for B dopant). The aforementioned doping concentrations are not limiting and other doping concentrations are within the spirit and the scope of this disclosure.

(55) In some embodiments, due to the base tapered profile of **1100** of fin structure **510** shown in FIG. **11**, the sidewall profile of S/D epitaxial structures **130** abutting fin structures **510** is not vertical. Instead, the sidewall profile of S/D epitaxial structures **130** abutting fin structures **510** follows the contour of fin structure **510** as shown in FIG. **12**. In some embodiments, the top width of S/D epitaxial structures **130** along the x-direction is larger than their bottom width along the same direction as shown, for example, in FIG. **12**.

(56) In some embodiments, after the formation of S/D epitaxial structures **130** on buried layer **110**, capping layer **705** and sacrificial gate structure **700** are removed to expose top and sidewall surfaces of fin structures **510**. An etching process selective to NS layers **505** (e.g., similar to the one used in operation **330** to etch the edge portions of NS layer **505**) removes NS layers **505** from fin structures **510** without removing NS layers **125** and spacer structures **135**. Subsequently the layers of gate structures **120** are formed on the exposed surfaces of NS layers **125** to form GAA-FETs **100** and **105** shown in FIG. **1**.

(57) In some embodiments, method **300** can be used to form buried layers **110** for n-type and p-type GAA-FETs on the same substrate. However, buried layer **110** should be doped with a suitable dopant type to achieve an appropriate p-n junction depletion region under each GAA-FET type. For example, for p-type GAA-FETs (e.g., with p-type doped S/D epitaxial structures), buried layer **110** should include As and/or P dopants (e.g., n-type dopants). Accordingly, for n-type GAA-FETs (e.g., with n-type doped S/D epitaxial structures), buried layer **110** should include boron dopants (e.g., p-type). In other words, buried layer **110** should include an opposite dopant type from that in S/D epitaxial structures **130**. In some embodiments, buried layers **110** formed as a wide bandgap material (e.g., Si:C) are suitable for both n-type and p-type GAA-FETs.

(58) According to some embodiments, FIGS. **13A**, **13B**, **13C**, and **13D** show combinations of n-type and p-type GAA-FETs with different types of buried layers formed on substrate **115** with method **300**. For example, FIG. **13A** shows p-type GAA-FETs formed on an n-type doped buried layer **110a** (e.g., doped with As and/or P), FIG. **13B** shows n-type GAA-FETs formed on a p-type doped buried layer **110b** (e.g., doped with B), and FIGS. **13C** and **13D** show p-type and n-type GAA-FETs formed on a wide bandgap buried layer **110c** (e.g., a Si:C buried layer). In some embodiments, any combination of p-type and n-type GAA-FETs with respective buried layers shown in FIGS. **13A**, **13B**, **13C**, and **13D** can be concurrently formed on substrate **115** using the operations of method **300**. By way of example and not limitation, this can be achieved by masking different areas of substrate **115** while other areas of substrate **115** are processed to form buried layers **110a**, **b**, and **c** with an appropriate type of transistor formed thereon. In some embodiments, the GAA-FETs and buried layers shown in FIGS. **13A**, **13B**, **13C**, and **13D** can be formed on the same area of substrate **115** or on different areas of substrate **115**.

(59) In some embodiments, the buried layers shown in FIGS. **13A**, **13B**, **13C**, and **13D** can be used with other transistor geometries, such as finFETs and GAA nano-wire FETs. Therefore, the buried layers shown in FIGS. **13A**, **13B**, **13C**, and **13D** are not limited to GAA-FETs with nano-sheet layers.

(60) The embodiments described herein are directed to methods for the fabrication of buried layers in GAA FETs, which prevent substrate etching and suppress junction leakage. In some embodiments, the buried layer is formed between the GAA FETs and the substrate at a thickness between about 5 nm and about 20 nm. According to some embodiments, the buried layers can be formed for p-type and n-type GAA-FETs and can include doped or un-doped (e.g., intrinsic) silicon layers. If un-doped, the buried layers include a wide bandgap material such as silicon with less than about 5 at. % carbon. If doped, the dopant type in the buried layer is opposite to that in the S/D

structures of the GAA-FETs. In some embodiments, the dopants are introduced during growth to achieve a uniform distribution within the buried layer and a high dopant concentration—e.g., higher than about 1×10^{21} atoms/cm³.

(61) In some embodiments, a structure includes a substrate and an epitaxial layer grown on the substrate, where the epitaxial layer includes a bandgap higher than that of intrinsic silicon. The structure further includes a transistor structure on the epitaxial layer.

(62) In some embodiments, a structure includes a substrate and a first epitaxial layer grown on the substrate, where the first epitaxial layer includes first dopants. The structure further includes a first transistor structure on the first epitaxial layer, where the first transistor structure includes source/drain (S/D) epitaxial structures with second dopants of an opposite type to that of the first dopants. Additionally, the structure includes a second epitaxial layer grown on the substrate, where the second epitaxial layer includes a bandgap higher than that of intrinsic silicon, and a second transistor structure on the second epitaxial layer, where the second transistor structure comprises S/D epitaxial structures with third dopants.

(63) In some embodiments, a method includes forming a doped epitaxial layer on a substrate, forming a stack of alternating first and second nano-sheet layers on the epitaxial layer, and patterning the stack and the epitaxial layer to form a fin structure. The method further includes forming a sacrificial gate structure on the fin structure, removing portions of the fin structure not covered by the sacrificial gate structure, and etching portions of the first nano-sheet layers. Additionally, the method includes forming spacer structures on the etched portions of the first nano-sheet layers, forming source/drain (S/D) epitaxial structures on the epitaxial layer abutting the second nano-sheet layers, where the S/D epitaxial structures comprise dopants of an opposite type to that of the epitaxial layer. The method also includes removing the sacrificial gate structure, removing the first nano-sheet layers, and forming a gate structure around the second nano-sheet layers.

(64) It is to be appreciated that the Detailed Description section, and not the Abstract of the Disclosure section, is intended to be used to interpret the claims. The Abstract of the Disclosure section may set forth one or more but not all possible embodiments of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the subjoined claims in any way.

(65) The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art will appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art will also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A structure, comprising: an epitaxial layer on a substrate and comprises a first dopant, wherein a band gap of the epitaxial layer is greater than a band gap of intrinsic silicon; a source/drain (S/D) region on the epitaxial layer and comprises a second dopant of a different type of the first dopant; and a gate structure in contact with a top surface of the epitaxial layer.
2. The structure of claim 1, wherein a dopant concentration of the epitaxial layer is greater than about 1×10^{21} atoms/cm³.
3. The structure of claim 1, wherein a carbon concentration of the epitaxial layer is between about 0 atomic percentage and about 5 atomic percentage.
4. The structure of claim 1, further comprising a channel region directly above the epitaxial layer.

5. The structure of claim 4, wherein the gate structure surrounds the channel region.
 6. The structure of claim 1, wherein the epitaxial layer comprises a depletion region between the S/D region and the substrate.
 7. The structure of claim 1, wherein gate structure comprises a gate electrode and a gate dielectric between the gate electrode and the epitaxial layer.
 8. A method, comprising: forming an epitaxial layer on a substrate, wherein the epitaxial layer is doped with dopants of a first type; forming a channel region on the epitaxial layer; forming a source/drain (S/D) region abutting the channel region, wherein the S/D region is doped with dopants of a second type opposite to the first type; and forming a gate structure surrounding the channel region, wherein the epitaxial layer is in contact with a bottom surface of the gate structure.
 9. The method of claim 8, wherein forming the S/D region comprises forming a p-n junction depletion region between the S/D region and the substrate.
 10. The method of claim 8, wherein forming the epitaxial layer comprises doping the epitaxial layer with carbon.
 11. The method of claim 8, wherein forming the epitaxial layer comprises forming the epitaxial layer with a bandgap wider than that of intrinsic silicon.
 12. The method of claim 8, wherein forming the gate structure comprises forming the gate structure on a top surface of the epitaxial layer.
 13. The method of claim 8, further comprising forming a spacer structure between the epitaxial layer and the channel region.
 14. The method of claim 13, wherein forming the spacer structure comprises forming a tapered profile of the spacer structure.
 15. A method, comprising: forming an epitaxial layer on a substrate by epitaxially growing a semiconductor material having a band gap greater than that of silicon; forming a gate-all-around field effect transistor (GAA-FET) on the epitaxial layer, wherein forming the GAA-FET comprises: forming channel regions on the epitaxial layer; forming a source/drain (S/D) epitaxial structure on the epitaxial layer and abutting the channel regions; and forming a gate structure surrounding the channel regions; and forming a depletion region between the GAA-FET and the substrate.
 16. The method of claim 15, wherein forming the depletion region comprises doping the epitaxial layer and the S/D epitaxial structure with opposite types of dopants.
 17. The method of claim 15, wherein forming the channel regions comprises forming a stack of nano-sheet layers on the epitaxial layer.
 18. The method of claim 15, wherein forming the epitaxial layer comprises depositing the epitaxial layer at a thickness between about 5 nm and about 20 nm.
 19. The method of claim 15, wherein forming the GAA-FET further comprises forming a spacer structure between the gate structure and the S/D epitaxial structure.
 20. The method of claim 19, wherein forming the spacer structure comprises forming the spacer structure on a top surface of the epitaxial layer.
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